

Vaidehi Patil

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RESEARCH INTERESTS

Language Models, AI safety, Privacy, Security, Multimodality, Multi-Agent privacy, Interpretability, Model editing

EDUCATION

University of North Carolina at Chapel Hill (UNC Chapel Hill) Aug 2022 - May 2027 (Expected)
PhD student in Computer Science
Graduate Research Assistant, Advisor: [Prof. Mohit Bansal](#)

Indian Institute of Technology Bombay (IIT Bombay) (CPI: 9.52/10) 2017 - 2022
Interdisciplinary Dual Degree: B. Tech. in Electrical Engineering, M. Tech. in AI and Data Science
Minor in Computer Science and Engineering
🏆 **Undergraduate Research Award, Best IDDDP Dissertation Award**
Thesis: Multilingual Representations for Closely Related Languages
Advisors: [Prof. Sunita Sarawagi](#), IIT Bombay; [Dr. Partha Talukar](#), Google Research India

WORK EXPERIENCE

Apple, Cupertino, USA May - Aug, 2024
ML Research Scientist Intern
Advisors: [Siddharth Patwardhan](#)
Improving faithfulness in RAG by reducing conflict between parametric and contextual knowledge

Amazon Alexa team, Seattle, USA May - October, 2023
Applied Scientist Intern
Advisors: [Markus Dreyer](#), [Mengwen Liu](#) & [Leonardo Ribeiro](#)
Self-refinement of multimodal LLMs to create a dataset for multimodal summarization

Adobe Research India May - August, 2021
Research Intern (🏆Pre-placement offer as Software Development Engineer)
Advisors: [Dr. Vishwa Vinay](#) & [Dr. Kuldeep Kulkarni](#)
Worked on scene expansion via scene graph using graph generative models

Adobe Research India May - August, 2020
Research Intern (🏆Re-internship offer as research intern for Summer 2021.)
Advisor: [Natwar Modani](#)
Introduced a novel problem of detecting versions of documents and created a dataset for it

AWL Japan February - April, 2022
Computer Vision R&D Intern

PUBLICATIONS

* - equal contribution

1. **UPCORE: Utility-Preserving Coreset Design for Balanced Unlearning.**
Vaidehi Patil, Elias Stengel-Eskin, Mohit Bansal
Under Review at ICML 2025
9. **Safree: Training-Free and Adaptive Guard for Safe Text-to-Image And Video Generation .**
Jaehong Yoon*, Shoubin Yu*, **Vaidehi Patil**, Huaxiu Yao, Mohit Bansal
ICLR 2025
8. **Unlearning Sensitive Information in Multimodal LLMs: Benchmark and Attack-Defense Evaluation.**
Vaidehi Patil, Yi-Lin Sung, Peter Hase, Jie Peng, Tianlong Chen, Mohit Bansal
TMLR 2024

7. **RefineSumm: Self-Refining MLLM for Generating a Multimodal Summarization Dataset**.
Vaidehi Patil, Leonardo F. R. Ribeiro, Mengwen Liu, Mohit Bansal and Markus Dreyer.
ACL Main Conference 2024 (🏆 Amazon Conference Travel Grant)
6. **Can Sensitive Information Be Deleted From LLMs? Objectives for Defending Against Extraction Attacks**.
Vaidehi Patil*, Peter Hase* and Mohit Bansal
ICLR 2024 as Spotlight
5. **Debiasing Multimodal Models via Causal Information Minimization**.
Vaidehi Patil, Adyasha Maharana and Mohit Bansal
EMNLP 2023 Findings
NeurIPS 2023 workshop on Causal Representation Learning
4. **GEMS: Scene Expansion using Generative Models of Graphs**.
Rishi Agarwal*, Tirupati Saketh Chandra*, **Vaidehi Patil***, Aniruddha Mahapatra*, Kuldeep Kulkarni and Vishwa Vinay.
WACV 2023
3. **Overlap-based Vocabulary Generation Improves Cross-lingual Transfer Among Related Languages**.
[Oral Presentation]
Vaidehi Patil, Partha Talukdar and Sunita Sarawagi.
ACL Main Conference 2022 (🏆 Google Conference Travel Grant)
2. **Detecting Document Versions and Their Ordering In a Collection**.
Natwar Modani, Anurag Maurya, Gaurav Verma, Inderjeet Nair, **Vaidehi Patil**, Anirudh Kanfode.
International Conference on Web Information Systems Engineering (WISE) 2021
🏆 **Best Paper Runner-Up Award**.
1. **Exploiting Language Relatedness for Low Web-Resource Language Model Adaptation: An Indic Languages Study**.
Yash Khemchandani*, Sarvesh Mehtani*, **Vaidehi Patil**, Abhijeet Awasthi, Partha Talukdar and Sunita Sarawagi.
ACL Main Conference 2021

Patents

2. **Expanding a scene graph using proposals from a generative model of scene graphs**.
Vishwa Vinay, Tirupati Saketh Chandra, Rishi Agarwal, Kuldeep Kulkarni, Hiransh Gupta, Aniruddha Mahapatra, **Vaidehi Patil**
US patent application | Adobe Inc.
1. **Systems for generating indications of relationships between electronic documents**.
Natwar Modani, **Vaidehi Patil**, Inderjeet Nair, Gaurav Verma, Anurag Maurya, Anirudh Kanfode.
US patent application | Adobe Inc.

PROFESSIONAL RESPONSIBILITIES

- *Lead Workshop Organizer* - **Machine Unlearning for Generative AI**, ICML 2025
- *Reviewer* - TMLR 2025, ICLR 2025, ACL 2025, NAACL 2025, TMLR 2024, ACL ARR 2024, CVPR 2022
- *Undergraduate TA* - Center for Machine Intelligence and Data Science, IIT Bombay

MAJOR ACADEMIC ACHIEVEMENTS

- Awarded **Carolina Computing Fellowship** by the Computer Science Dept, UNC Chapel Hill [’22]
- **2nd rank** in AI and Data Science batch of Centre for Machine Intelligence and Data Science [’22]
- Granted Advanced Performers grade (awarded to **top 1%**) grade for excellent performance [’18]
- Awarded an option of branch change due to exceptional academic performance in first year [’18]
- Secured All India Rank in **top 0.25%** in JEE Mains 2017 and **top 2%** in JEE Advanced 2017 [’17]

INVITED TALKS

- **Deep Learning: Classics and Trends**
Can Sensitive Information Be Deleted from LLMs? [Sep ’24]
- **Ploutos**
UPCORE: Utility-Preserving Coreset Selection for Balanced Unlearning [Apr ’25]